# ZHANG, Shu-yang, PhD

**CV | 2012-2026**

Research Fellow at City Syntax Lab, National University of Singapore

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## RESEARCH INTERESTS

- Urban building energy model (UBEM)
- Urban climate
- Mobility and heat risk
- Inequity in heatwave
- Shading structure construction

## SKILLS

- Coding: Python (Machine learning, Deep learning)
- Geography and Remote Sensing: ArcGIS, QGIS (PyQGIS, UMEP, SOLWEIG_GPU), ENVI, SNAP
- Building Simulation: DesignBuilder, Ecotect, Radiance, Daysim, Dpethmap
- 3D Modeling: Rhino + Grasshopper (Ladybug, Honeybee, Dragonfly, Eddy3D), SketchUp
- 2D Graphic: AutoCAD, Photoshop, InDesign, Illustrator, PowerPoint
- Statistics: Excel, SPSS, Minitab, Origin
- Rendering: Enscape, V-ray
- Video Editing: Premiere, Audition

## EDUCATION

- 2020.08 – 2025.09 **School of Architecture, Tsinghua University** Beijing, China
  *Ph.D. in Engineering (Supervisor: Prof. Nian-xiong Liu)*
  *Dissertation: Design Optimization of Outdoor Thermal Environment and Building Form Layout under Extreme Climate Risks*
- 2023.10 – 2024.10 **Department of Architecture, National University of Singapore** Singapore
  *Visiting Ph.D. student of Architecture (Supervisor: Associate Prof. Chao Yuan)*
- 2017.08 – 2020.06 **School of Architecture, South China University of Technology** Guangzhou, China
  *M.Arch of Architecture (Supervisor: Prof. Yang Wang)*
  *Dissertation: Research on Space and Morphology Design of Umbrella Structures Building*
- 2012.08 – 2017.06 **School of Architecture and Art, Hefei University of Technology** Hefei, China
  *B.arch of Architecture*

**LIST OF ACADEMIC PUBLICATIONS**

(† co-first author; * Corresponding Author)

ORCID: 0000-0002-3677-1363
Scopus Author ID: 57349228300
Web of Science Researcher ID: OIR-4914-2025
ResearchGate: https://www.researchgate.net/profile/Shuyang-Zhang-17/research
Google Scholar: https://scholar.google.com/citations?user=GW_6fuUAAAAJ (H-index: 8, Total citations: 195)

• **Publications in Refereed Journals** (published 18 papers, with 10 of them as the first author):

1. **Zhang S**, Chen T, Zhang X, Gao X, Chao Y, Liu N*. (2026). Local UHI mitigation and utilization: UBEM driven by localized weather data for addressing urban building energy risks under compound heatwave and UHI: a campus-scale study. ***Building and Environment***, 305, 115168. (SCI Q1, IF=8.4) https://doi.org/10.1016/j.buildenv.2026.115168
2. **Zhang S**, Wu X, Xu P, Liu N*, Li N*. (2026). Local UHI mitigation and utilization: Urban building energy modeling, simulation, and urban design responses based on localized weather data. ***Building Simulation***, 19, 859–883. (SCI Q1, IF=6.9) https://doi.org/10.1007/s12273-026-1410-7
3. Xu P†, Zhao P†, Xia B, Zhao Z, Liu N*, **Zhang S***, Du J*. (2025). Reciprocal frame design for large-scale timber construction. ***Nature communications***, 16, 11482. (SCI Q1, IF=18.1, the article is cited in ***Nature Reviews Materials***) https://doi.org/10.1038/s41467-025-66491-4
4. **Zhang S**, Liu N*, Wu X, Han Y, Li W, Guo F, Song X, Yuan C. (2025). An empirical study on the compound effects of extreme weather and UHI on building energy consumption under urban local climate. ***Building and Environment***, 287, 113824. (SCI Q1, IF=8.4) https://doi.org/10.1016/j.buildenv.2025.113824
5. **Zhang S**, Yuan C, Chen T, Ma B, Liu N*. (2024). A cross-scale indicator framework for the study of annual stability of land surface temperature in different land uses. ***Sustainable Cities and Society***, *116*, 105936. (SCI Q1, IF=13.3) https://doi.org/10.1016/j.scs.2024.105936
6. **Zhang S**, Yuan C, Ma B, Liu N*, Li W. (2024). Coupling effects of building-vegetation-land on seasonal land surface temperature on street-level: A study from a campus in Beijing. ***Building and Environment***, 262, 111790. (SCI Q1, IF=8.4) https://doi.org/10.1016/j.buildenv.2024.111790
7. **Zhang S**, Liu N*, Ma B, Yan S. (2023). The effects of street environment features on road running: An analysis using crowdsourced fitness tracker data and machine learning. ***Environment and planning B: Urban analytics and city science***, 51(2), 529-545. (SSCI Q1, IF=2.9) https://doi.org/10.1177/23998083231185589
8. **Zhang S**, Liu N*, Li W, Yan S. (2022). Umbrella structure building design method via case-based design and statistical analysis of structural morphological parameters. ***Journal of Building Engineering***, 45, 103542. (SCI Q1, IF=8.1, the article is cited in ***Nature Structural & Molecular Biology*** *and* ***Nature communications***) https://doi.org/10.1016/j.jobe.2021.103542
9. **Zhang S**, Wang Y*. (2020). Balancing of form and force: research on architectural design strategies based on application of eccentric umbrella structure. ***The Architect***, 2020(03), 91-99 (in Chinese, Tier 2 Chinese National Core Journal in Building Science) https://d.wanfangdata.com.cn/periodical/jzs202003013
10. **Zhang S**, Wang Y*. (2020). The historical evolution and contemporary value of umbrella structure: cases study based on the practice in Western Europe and Latin America since the 20th Century. ***South Architecture***, 2020(06), 22-29 (in Chinese, Tier 3 Chinese National Core Journal in Building Science, 7 citations) https://d.wanfangdata.com.cn/periodical/nfjz202006004
11. **Zhang S**, Wang Y*.(2019). Dynamic exterior sunshade design and shading effect analysis of southeast and southwest facing buildings in Guangzhou. ***Building Energy Efficiency***, 47(08), 91-95+100 (in Chinese, Tier 3 Chinese National Core Journal in Building Science, 13 citations) https://d.wanfangdata.

com.cn/periodical/fcyyy201908022

12. Wang G, Zhang X, **Zhang S**, Wang X, Chen T, Heng CK. (2026). The rural-urban gradient modulates ecosystem health's impact on environmental factors: evidence from Middle Yangtze. ***Stochastic Environmental Research and Risk Assessment***, 40, 204. (SCI Q1, IF= 3.8) https://doi.org/10.1007/s00477-026-03341-9
13. Xu P, Zhao P, **Zhang S**, Xu Z*. (2025). An intelligent construction system for timber structures without human intervention a case study of Chinese timber reciprocal structures. ***Architectural Journal***, 2025(11), 107-114 (in Chinese, Tier 1 Chinese National Core Journal in Building Science) https://doi.org/10.19819/j.cnki.ISSN0529-1399.202511014
14. Chen T, Singh V K, **Zhang S**, Zhang L, Yuan C*. (2025). A city-scale mapping tool for assessing effects of urban greenery and morphologies on thermal comfort: A case study in Singapore. ***Building and Environment***, 113760. (SCI Q1, IF=8.4) https://doi.org/10.1016/j.buildenv.2025.113760
15. Zhang X, Chen T*, **Zhang S**, Yuan C, PY Tan. (2025). Toward healthy neighborhood thermal environments: Interpretation of thermal comfort and rapid identification of hotspot areas using image-based metrics and explainable machine learning. ***Sustainable Cities and Society***, 106648 (SCI Q1, IF=13.3) https://doi.org/10.1016/j.scs.2025.106648
16. Li W, Zhou Z*, **Zhang S**, Feng J. (2022). Study on relationship between shading and outdoor air temperature based on the comparison of two high-rise residential estates with field measurements. ***Buildings***, 12(11), 1813. (SCI Q2, IF= 3.4) https://doi.org/10.3390/buildings12111813
17. Gao X, Yuan M, Yang Y, **Zhang S**, Ni J, Zhuang W, Geng Y*. (2026). Decoding occupant behavior in shared spaces of open-plan offices: A multi-sensor data fusion and machine learning approach. ***Building and Environment***, 114281. (SCI Q1, IF=8.4) https://doi.org/10.1016/j.buildenv.2026.114281
18. Wang Y, Lin M, Xu K, **Zhang S**, Ma H*. (2021). Energy consumption analysis of glass house using electrochromic window in the subtropical region. ***Journal of Engineering, Design and Technology***, 19(1), 203-218. (SCI Q1, IF= 2.7, the article is cited in ***Cell Reports Physical Science***) https://doi.org/10.1108/JEDT-12-2019-0348

**• Conference papers:**

1. **Zhang S**, Ang Y*. (2026). A future climate-responsive UBEM framework for building energy resilience under compound extreme heatwaves and UHI effects: a case study of Singapore. In *ASim 2026*, Singapore.

**• Theses:**

1. **Zhang S**. (2025). Design Optimization of Outdoor Thermal Environment and Building Form Layout under Extreme Climate Risks. ***Tsinghua University***, PhD thesis, (Nomination for the Excellent Doctoral Dissertation Award, Supervisor: Prof. Nian-xiong Liu)
2. **Zhang S**. (2020). Research on Spatial Morphology Design of Umbrella Structures Building. ***South China University of Technology***, M.Arch thesis, (Supervisor: Prof. Yang Wang)

**• Book Chapters:**

1. Liu N et al. (2021). A data supported object and performance-based design method for green dwellings. ***Beijing: Tsinghua University Press*** (in Chinese, ISBN 978-7-302-57312-8, **Authored Chapter 2**)
2. Zhang Y, Liu N, Cheng X et al. (2025). Design Theories and Indicator Framework for High-Quality Green Building. ***Beijing: China Architecture Publishing Press*** (in Chinese, ISBN 978-7-112-31625-0, **Co-authored Chapter 6**)

## PATENTS

**• Granted Patents:**

1. Hao C, **Zhang S**, Shi S, Liu S, Zhao S, Song Y, Chu Y. (2025). High airtight flush doors with replaceable thresholds and construction methods. China: CN113914746A. (Invention patents)
2. Liu N, **Zhang S**, Shi S, Lin B. (2026). Calculation method and device of urban building energy consumption based on localized weather data. China: CN121615206B. (Invention patents)

**• Patents Pending:**

1. Liu N, **Zhang S**, Wu X. A method for calculating heat exposure risk considering urban mobility flows and built environment dynamics. (Invention patents, originally submitted in July 2025)
2. Liu N, **Zhang S**. A method and device for calculating the height and canopy distribution of urban evergreen and deciduous trees based on GF-7 satellite imagery. (Invention patents, originally submitted in July 2025)

## AWARDS AND GRANTS

| | |
|---|---|
| • 2025 | **Merit Graduate (University Level)**<br>**Nomination for the Excellent Doctoral Dissertation Award (School Level)**<br>*Tsinghua University, School of Architecture* |
| • 2023 | **National Postgraduate Scholarship-Joint PhD Training Program**<br>*China Scholarship Council, Grant Amount: SGD 26,400, Grant No. 202306210331* |
| • 2017 | **Graduate Freshman Scholarship**<br>*South China University of Technology* |
| • 2017 | **Outstanding Bachelor's Graduate**<br>*Hefei University of Technology* |

## RESEARCH PROJECTS

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|---|---|
| • 2026.03- | **Using physics-based and AI-driven urban building energy modeling (UBEM) for operational and embodied carbon emissions estimation in a seed region, Supervisor as PI** (Principal Researchers)<br>*Funding amount: SDG 714,000, Grant No. SC-URA-NUS-2025-1* |
| • 2022-2025 | **The National Key R&D Program of China, Supervisor as Co-PI** (Subproject coordinator, Principal Researchers)<br>*Funding amount: CNY 3,200,000, Grant No.2022YFC3803801* |
| • 2021-2025 | **The National Natural Science Foundation of China (Key Program), Supervisor as Co-PI** (Participation)<br>*Funding amount: CNY 3,000,000, Grant No.52130803* |
| • 2020-2025 | **The National Natural Science Foundation of China (General Program), Supervisor as PI** (Participation)<br>*Funding amount: CNY 580,000, Grant No.52078262* |
| • 2016-2020 | **The National Key R&D Program of China, Supervisor as PI** (Participation)<br>*Funding amount: CNY 2,690,000, Grant No.2016YFC0700206* |

## ARCHITECTURE DESIGN

| | | |
|---|---|---|
| • 2021.02 - 2021.03 | **Ceramic cultural and creative park**<br>*Plan drawing with CAD* | Zibo, Shandong, China |

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|---|---|---|
| • 2019.06 - 2019.07 | **Gymnasium, Yihua Experimental School**<br>*Schematic design* | Yueyang, Hunan, China |
| • 2019.05 - 2019.06 | **New campus planning for the Vocational Institute of Ceramic Arts and Crafts (built)**<br>*Campus planning, training and laboratory center design, dormitory design* | Jingdezhen, Jiangxi, China |
| • 2019.03 - 2019.05 | **Urban design competition for the Core Area of the High-speed Rail Business District (3rd Prize)**<br>*Concept design and southern building cluster planning* | Jingdezhen, Jiangxi, China |
| • 2018.11 - 2018.12 | **Integrated practical training center，Vocational School of Health and Medicine (built)**<br>*Schematic design* | Yantai, Shandong, China |
| • 2018.05 - 2019.07 | **Foreign Languages Vocational College (built)**<br>*Detailed construction plan, sunlight analysis report* | Rizhao, Shandong, China |
| • 2017.11 - 2018.01 | **Jilin University - Zhuhai campus (built)**<br>*Specialist apartments and national academy building design* | Zhuhai, Guangdong, China |

## DESIGN AWARDS

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|---|---|---|
| • 2017 | **i.FUTURE 1st "Future Cities" International Concept Design Competition**<br>*China Institute of Urbanization, Zhejiang University* | 2nd Prize (2/2)<br>*Award: CNY 5,000* |
| • 2016 | **The 6th IDEA-KING International Landscape Planning & Design Competition**<br>*International Landscape Design Industry Association* | The Excellence Award (3/4) |
| • 2015 | **Shekou Prince Bay Public Cultural Architecture Design Competition**<br>*Young Bird Plan, China* | Top 50 (1/2) |
| • 2014 | **National Green Building Design Competition, Light Green Category**<br>*China Green Building Association* | 3rd Prize (2/3) |

## PRESENTATIONS

**• Seminar**

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|---|---|
| • 2024.02 | **Earth observation and urban morphometrics for urban microclimate research**<br>*Urban Climate Design Lab, Department of Architecture, National University of Singapore* |
| • 2026.03 | **Empirical analysis and application of UCM and UBEM tools based on localized weather data**<br>*City Syntax Lab, Department of Built Environment, National University of Singapore* |

## TEACHING EXPERIENCE

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|---|---|
| • 2022.02-2022.06 | **Teaching Assistant**, "***Building Physics - Thermal Environment***" (junior undergraduate professional theory course), **Tsinghua University**<br>*Assisted with teaching, supervised experiments, and graded assignments (TA stipend provided)* |

• 2018.09-2018.12 **Teaching Assistant**, "***Expert Studio Design***" **and** "***Youth Apartment Design***" (second-year undergraduate professional design course), **South China University of Technology**
*Assist in design guidance, calculate grades, and scan design drawings for archiving*

## REVIEWER ASSIGNMENTS

*Sustainable Cities and Society,*
*Cities,*
*Environment and Planning B: Urban Analytics and City Science*
*Building Simulation 2027(Conference)*

## PROFESSIONAL AFFILIATIONS

• Since 2022 **The Architectural Society of China** (Member)

## WORKSHOP PARTICIPATION

• 2017.11 - 2017.12 **The Skycourt and Skygarden: Towards a Vertical City Theory**
*School of Architecture, South China University of Technology*

*Oct 16, 2026*